



NDA 220605

NDA APPROVAL

Takeda Pharmaceuticals America, Inc.
Attention: Sarah Kim
Associate Director, Global Regulatory Affairs
40 Landsdowne Street
Cambridge, MA 02139

Dear Sarah Kim:

Please refer to your new drug application (NDA) received December 29, 2025, and your amendments, submitted under section 505(b) of the Federal Food, Drug, and Cosmetic Act (FD&C Act) for Mimrylo (rusfertide) for injection.

This NDA provides for the use of Mimrylo (rusfertide) for injection for the treatment of erythrocytosis in adults with polycythemia vera (PV).

APPROVAL & LABELING

We have completed our review of this application, as amended. It is approved, effective on the date of this letter, for use as recommended in the enclosed agreed-upon labeling.

CONTENT OF LABELING

As soon as possible, but no later than 14 days from the date of this letter, submit the content of labeling [21 CFR 314.50(l)] in structured product labeling (SPL) format using the FDA automated drug registration and listing system (eLIST), as described at FDA.gov.¹ Content of labeling must be identical to the enclosed labeling (text for the Prescribing Information, Patient Package Insert, and Instructions for Use) as well as annual reportable changes not included in the enclosed labeling. Information on submitting SPL files using eLIST may be found in guidance for industry *SPL Standard for Content of Labeling Technical Qs and As*.²

The SPL will be accessible via publicly available labeling repositories.

¹ <http://www.fda.gov/ForIndustry/DataStandards/StructuredProductLabeling/default.htm>

² We update guidances periodically. For the most recent version of a guidance, check the FDA Guidance Documents Database <https://www.fda.gov/RegulatoryInformation/Guidances/default.htm>.

CARTON AND CONTAINER LABELING

Submit final printed carton and container labeling that are identical to the enclosed carton and container labeling, as soon as they are available, but no more than 30 days after they are printed. Please submit these labeling electronically according to guidance for industry *SPL Standard for Content of Labeling Technical Qs & As*. For administrative purposes, designate this submission “**Final Printed Carton and Container Labeling for approved NDA 220605.**” Approval of this submission by FDA is not required before the labeling is used.

DATING PERIOD

Based on the stability data submitted to date, the expiry dating period for Mimrylo (rusfertide) for injection shall be 24 months from the date of manufacture when stored at 20°C to 25°C.

ADVISORY COMMITTEE

Your application for Mimrylo was not referred to an FDA advisory committee because the application did not raise significant public health questions on the role of the drug in the diagnosis, cure, mitigation, treatment, or prevention of a disease.

REQUIRED PEDIATRIC ASSESSMENTS

Under the Pediatric Research Equity Act (PREA) (21 U.S.C. 355c), all applications for new active ingredients (which includes new salts and new fixed combinations), new indications, new dosage forms, new dosing regimens, or new routes of administration are required to contain an assessment of the safety and effectiveness of the product for the claimed indication in pediatric patients unless this requirement is waived, deferred, or inapplicable.

Because this drug product for this indication has an orphan drug designation, you are exempt from this requirement.

POSTMARKETING REQUIREMENTS UNDER 505(o)

Section 505(o)(3) of the FD&C Act authorizes FDA to require holders of approved drug and biological product applications to conduct postmarketing studies and clinical trials for certain purposes, if FDA makes certain findings required by the statute.

We have determined that an analysis of spontaneous postmarketing adverse events reported under subsection 505(k)(1) of the FD&C Act will not be sufficient to assess a signal of serious risk of thrombocytosis, and to identify an unexpected serious risk of thromboembolic events, hematologic transformation to myelofibrosis or acute myeloid leukemia, and increase in JAK2 allele burden.

Furthermore, the active postmarket risk identification and analysis system as available under section 505(k)(3) of the FD&C Act will not be sufficient to assess these serious risks.

Finally, we have determined that only a clinical trial (rather than a nonclinical or observational study) will be sufficient to assess a signal of serious risk of thrombocytosis and to identify an unexpected serious risk of thromboembolic events, hematologic transformation to myelofibrosis or acute myeloid leukemia, and increase in JAK2 allele burden.

Therefore, based on appropriate scientific data, FDA has determined that you are required to conduct the following trial:

- 5050-1 Conduct a study to characterize the long-term safety of rusfertide in patients with polycythemia vera based on a pooled analyses of data from studies PTG-300-11 (VERIFY) and PTG-300-04 (REVIVE) / PTG-300-21 (THRIVE). Provide at least 3 years of follow-up for enrolled subjects treated with the rusfertide dose titration regimen. Safety outcomes of interest include thrombocytosis, thromboembolic events, hematologic transformation to myelofibrosis or acute myeloid leukemia, JAK2 allele burden, and cytoreductive therapy use.

Include a summary of the major safety findings for all patients and submit datasets with the final clinical study report submission.

The timetable you submitted on August 17, 2026, states that you will conduct this trial according to the following schedule:

Final Report Submission: 06/2029

FDA considers the term *final* to mean that the applicant has submitted a protocol, the FDA review team has sent comments to the applicant, and the protocol has been revised as needed to meet the goal of the study or clinical trial.³

Submit clinical protocol(s) to your IND 152414 with a cross-reference letter to this NDA. Submit nonclinical and chemistry, manufacturing, and controls protocols and all final report(s) to your NDA. Prominently identify the submission with the following wording in bold capital letters at the top of the first page of the submission, as appropriate:

REQUIRED POSTMARKETING PROTOCOL UNDER 505(o), REQUIRED POSTMARKETING FINAL REPORT UNDER 505(o), REQUIRED POSTMARKETING CORRESPONDENCE UNDER 505(o).

³ See the guidance for Industry *Postmarketing Studies and Clinical Trials—Implementation of Section 505(o)(3) of the Federal Food, Drug, and Cosmetic Act* (October 2019).
<https://www.fda.gov/RegulatoryInformation/Guidances/default.htm>.

Section 505(o)(3)(E)(ii) of the FD&C Act requires you to report periodically on the status of any study or clinical trial required under this section. This section also requires you to periodically report to FDA on the status of any study or clinical trial otherwise undertaken to investigate a safety issue. Section 506B(a)(1) of the FD&C Act, as well as 21 CFR 314.81(b)(2)(vii) requires you to report annually on the status of any postmarketing commitments or required studies or clinical trials.

FDA will consider the submission of your annual report under section 506B(a)(1) and 21 CFR 314.81(b)(2)(vii) to satisfy the periodic reporting requirement under section 505(o)(3)(E)(ii) provided that you include the elements listed in 505(o) and 21 CFR 314.81(b)(2)(vii). We remind you that to comply with 505(o), your annual report must also include a report on the status of any study or clinical trial otherwise undertaken to investigate a safety issue. Failure to submit an annual report for studies or clinical trials required under 505(o) on the date required will be considered a violation of FD&C Act section 505(o)(3)(E)(ii) and could result in enforcement action.

POSTMARKETING COMMITMENTS SUBJECT TO REPORTING REQUIREMENTS UNDER SECTION 506B

We remind you of your postmarketing commitments:

- 5050-2 Complete Study PTG-300-11 (VERIFY): A Phase 3 Study of the Hepcidin Mimetic Rusfertide (PTG-300) in Patients with Polycythemia Vera.

The timetable you submitted on August 17, 2026, states that you will conduct this study according to the following schedule:

Study/Trial Completion: 08/2028
Final Report Submission: 12/2028

- 5050-3 Complete Study PTG-300-21, an ongoing multicenter, open-label, phase 3 trial to assess long-term safety, tolerability, and efficacy of treatment with rusfertide.

The timetable you submitted on August 17, 2026, states that you will conduct this study according to the following schedule:

Study/Trial Completion: 01/2027
Final Report Submission: 04/2027

Submit clinical protocols to your IND 152414 for this product. Submit nonclinical and chemistry, manufacturing, and controls protocols and all postmarketing final reports to this NDA. In addition, under 21 CFR 314.81(b)(2)(vii) and 314.81(b)(2)(viii) you should include a status summary of each commitment in your annual report to this NDA. The status summary should include expected summary completion and final report submission dates, any changes in plans since the last annual report, and, for clinical

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studies/trials, number of patients/subjects entered into each study/trial. All submissions, including supplements, relating to these postmarketing commitments should be prominently labeled “**Postmarketing Commitment Protocol**,” “**Postmarketing Commitment Final Report**,” or “**Postmarketing Commitment Correspondence**.”

PROMOTIONAL MATERIALS

You may request advisory comments on proposed introductory advertising and promotional labeling. For information about submitting promotional materials, see guidance for industry *Providing Regulatory Submissions in Electronic and Non-Electronic Format—Promotional Labeling and Advertising Materials for Human Prescription Drugs*.

As required under 21 CFR 314.81(b)(3)(i), you must submit final promotional materials, and the Prescribing Information, at the time of initial dissemination or publication, accompanied by a Form FDA 2253. Form FDA 2253 is available at FDA.gov.⁴ Information and Instructions for completing the form can be found at FDA.gov.⁵

REPORTING REQUIREMENTS

We remind you that you must comply with reporting requirements for an approved NDA (21 CFR 314.80 and 314.81).

REQUESTED ENHANCED PHARMACOVIGILANCE (EPV)

We request that for Mimrylo you submit all serious domestic and foreign cases of the following events as though they were 15-day Alert reports (which are described under 21 CFR 314.80(c)(1)) through the fifth year following initial U.S. approval:

- New onset or worsening thrombocytosis
- Hematologic transformation to myelofibrosis or acute myeloid leukemia
- Arterial or venous thromboembolic events
- Janus kinase 2 allele burden increase

We request that you investigate all such reports of serious events as though they were subject to 21 CFR 314.80(c)(1)(ii). When you receive new information for a previously reported case, submit a follow-up ICSR using the same unique case identification number (as described in 21 CFR 314.80(f)(5)(viii)).

Include a narrative summary including analyses of the above events reported with Mimrylo as part of your required periodic safety reports (e.g., periodic adverse drug experience report (PADER) required under 21 CFR 314.80(c)(2)), quarterly during the

⁴ <http://www.fda.gov/downloads/AboutFDA/ReportsManualsForms/Forms/UCM083570.pdf>

⁵ <http://www.fda.gov/downloads/AboutFDA/ReportsManualsForms/Forms/UCM375154.pdf>

first 3 years postapproval and annually thereafter, through the 5th year following initial U.S. approval date.

Your analyses should include interval and cumulative data from the date of approval of Mimrylo including cases from your global safety, and clinical trials databases. Your analyses should provide an assessment of causality for each case, and document the following, where available:

- Indication for use (including all labeled and off-label use)
- Temporal association
- Dose and duration of therapy
- Associated signs and symptoms
- Confounders and underlying risk factors
- Concomitant medications, including cytoreductive therapy use and dose
- Treatment given for the event
- Outcome
- Dechallenge/rechallenge information

POST APPROVAL FEEDBACK MEETING

New molecular entities qualify for a post approval feedback meeting. Such meetings are used to discuss the quality of the application and to evaluate the communication process during drug development and marketing application review. The purpose is to learn from successful aspects of the review process and to identify areas that could benefit from improvement. If you would like to have such a meeting with us, call the Regulatory Project Manager for this application.

COMPENDIAL STANDARDS

A drug with a name recognized in the official United States Pharmacopeia or official National Formulary (USP-NF) generally must comply with the compendial standards for strength, quality, and purity, unless the difference in strength, quality, or purity is plainly stated on its label (see FD&C Act § 501(b), 21 USC 351(b)). FDA typically cannot share application-specific information contained in submitted regulatory filings with third parties, which includes USP-NF. To help ensure that a drug continues to comply with compendial standards, application holders may work directly with USP-NF to revise official USP monographs. More information on the USP-NF is available on USP's website⁶.

⁶ <https://www.uspnf.com/>

If you have any questions, contact Carleveva Thompson, Regulatory Project Manager, at 301-796-1403 or Carleveva.Thompson@fda.hhs.gov.

Sincerely,

{See appended electronic signature page}

Lisa Yanoff, MD
Deputy Director for Clinical Science
Center for Drug Evaluation and Research

ENCLOSURES:

- Content of Labeling
 - Prescribing Information
 - Patient Package Insert
 - Instructions for Use
- Carton and Container Labeling

This is a representation of an electronic record that was signed electronically. Following this are manifestations of any and all electronic signatures for this electronic record.

/s/

LISA B YANOFF
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